IP Camera Module — Speaker Script

Final presentation · Part 01 · slides 05–15 · about 5½–6 minutes

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The logic in one sentence

Everything in this part answers one question — how does the camera decide what to record? A useful clip must meet two requirements: (1) it includes the start of the movement, and (2) it is about the gecko, not people or lights. The setup comes first, then one section for each requirement, then results.

Structure

Slides	Section	Key message
05	**The question**	How does the camera decide what to record?
06–07	**Setup**	PoE camera → ROCK 4C+ recorder; main stream recorded, substream analysed
08	**Requirement 1 · the start**	Never stop recording: keep 4 s before and 5 s after the motion
09–12	**Requirement 2 · only the gecko**	Three stages: any change → detection area → the gecko itself
13–14	**How we find the gecko**	Colour, not brightness, separates the lizard from the desk
15	**Results & next**	Both requirements met in the lab; a real gecko is next

Three numbers to remember

4–6 s
kept before the first motion

86 → 0
unwanted clips: stage 1 at school → hand-only clips in round 3

0.23 s
delay from camera to ROCK

How to read this script: the green line under each title is the one point the audience must remember from that slide. Text in ***[brackets]*** is a cue, not to be read aloud. Arrows (→) mark the sentence that links to the next section.

Script

Slide 05 IP Camera Module ≈20 s

> KEY POINT **One question: how does the camera decide what to record?**

I'm Kecheng, and I'll present the IP camera module. Everything in this part answers one question: how does the camera decide what to record?

A useful clip has to meet two requirements. First, it must include the start of the movement. Second, it must be about the gecko — not about people or lights. I'll show the setup, then how we meet each requirement, and finish with results and limits.

Slide 06 Hardware Connection & Power ≈30 s

> KEY POINT **One cable powers the camera; the ROCK 4C+ is the recorder.**

This is our real setup. The camera gets 48-volt power and sends video over one network cable from the PoE injector. The injector, the ROCK 4C+ and the MacBook meet at a gigabit switch. The ROCK is the recorder; the Mac keeps the finished clips.

One weak point: ***[point to the right USB-C]*** the ROCK has no adapter of its own — the Mac's USB-C port powers it. The ROCK stopped suddenly twice, and we suspect this power path, but it is not confirmed.

Slide 07 Two Streams, One Clip per Event ≈35 s

> KEY POINT **The main stream is recorded; the small substream is analysed.**

The camera sends two streams, and we give them different jobs. The full 2560-by-1440 main stream is only copied, never re-encoded, so recording costs almost nothing. The small substream is shrunk to 160 by 90 and analysed to decide when an event starts and ends. Then the clip is cut, checked with SHA-256 and moved to the Mac.

[Point to the latency test] We measured the delay by filming a clock on the Mac screen: 0.23 seconds from camera to ROCK. Detection and tracking use about 13 percent of the ROCK's CPU, so there is plenty of headroom.

→ With this setup, the first requirement: the start of the movement.

Slide 08 Keep the Seconds Before Motion

≈40 s

> KEY POINT **Requirement 1: never stop recording — keep 4 s before and 5 s after the motion.**

If we start recording only after motion is detected, the start is always lost: detection takes up to 0.65 seconds, connecting takes 0.4, and FFmpeg needs about 2 seconds before it writes anything.

So our recorder never stops. It writes 2-second segments into a 60-second ring in RAM. When an event ends, we join the segments from 4 seconds before the first motion to 5 seconds after the last one, without re-encoding. ***[Point to the clip]*** In this real clip the lizard is pulled at 5.1 seconds, and you can still see it before it moved. 97 clips have been saved and checked this way.

→ That solves the start. The second requirement is harder: record only the gecko.

Slide 09 Stage 1 · Any Change

≈25 s

> KEY POINT **Counting every changed pixel records far too much: 86 clips in 2.5 h.**

We reached it in three stages. Stage one counts every pixel that changes between frames. It is fast, but it cannot tell what moved. At school it made 86 recordings in two and a half hours, with no gecko in the scene — people, hands, lights and auto-exposure all counted as motion.

Slide 10 Stage 2 · Detection Area

≈30 s

> KEY POINT **Watch only the enclosure, and ignore whole-picture changes.**

Stage two tells it where to look. We draw the enclosure on the web page, and changes outside it are ignored. If 30 percent or more of the picture changes at once, it is a lighting or exposure change, not motion. And motion must last half a second.

Result: zero triggers from outside the area, and a 31.4 percent whole-picture change was ignored. But one problem is left: ***[point to the third frame]*** a hand inside the area still records.

Slide 11 Stage 3 · Lock on the Gecko

≈30 s

> KEY POINT **The trigger is the gecko itself, not any change.**

So in stage three, the trigger is the gecko itself. A clip starts only when the tracked gecko really moves: its whole box must shift at least 2 pixels within a second, with both edges moving the same way. A shadow only stretches one edge, so it does not count.

The old area rule still runs, but only writes to the log, so we can compare the two rules on the same video.

Slide 12 Stage 3 · On-site Tests

≈35 s

> KEY POINT **Three lab rounds; hand-only clips went 3 → 2 → 0.**

We tested stage three in the lab in three rounds, and each failure became a rule. Round one: three clips with only a hand — pixel flicker looked like movement, so we required real travel for half a second. Round two: two — the hand's shadow joined the gecko and pulled the box, so we added the whole-box shift, an overlap check and a size limit.

Round three: zero. A hand stayed in the area for 8 seconds, the old rule fired on 87 frames, nothing was recorded, and all three gecko clips were kept.

→ All of this depends on knowing where the gecko is. Here is how we find it.

Slide 13 Gecko Tracking · Method

≈30 s

> KEY POINT **Colour, not brightness, separates the lizard from the desk.**

The tracker learns a background from still frames, finds the pixels that differ from it, groups them into blobs, and locks on the one that moves.

The key was colour. In brightness, our green lizard is almost the same grey as the desk — 157 against 156 to 186 — so at first the box covered only its tail. In the colour channels they are far apart. The tracker runs in pure Python at about 8 milliseconds per frame.

Slide 14 Gecko Tracking · Results

≈20 s

> KEY POINT **The box stays on the lizard while it moves and while it rests.**

With colour, the box covers the whole lizard in 5 of 6 sampled frames, and it stayed on the lizard through 55 seconds of rest without a false "lost". The limit: if the lizard is pulled away fast, the box can stay on the empty spot for about 2 seconds.

Slide 15 Results, Limits & Next Steps

≈35 s

> KEY POINT **Both requirements are met in the lab; a real gecko is the next test.**

To sum up: both requirements are met in the lab. Clips keep 4 to 6 seconds before the first motion, and changes outside the area, lighting changes and a hand in the area no longer start a recording.

What we have not tested yet: a real gecko — every test used a toy lizard — night recording with the camera's infrared, and runs longer than a few hours. Next, we test in the real enclosure, run a night test, give the ROCK its own power supply for a 24-hour run, and link clips to thermal events.

→ Which brings us to the thermal sensor. *[Hand over to the next speaker]*

If time is short (about 4 minutes)

- Slide 07: say only the two jobs of the two streams and 0.23 s; skip the CPU figure.
- Slides 09 and 10: one sentence each — “86 recordings in 2.5 h” and “outside the area is ignored, but a hand inside still records”.
- Slide 12: say only “3 → 2 → 0 hand-only clips”.
- Slide 13: say only the colour point; skip the five steps.

If someone asks

Why not an ESP32-C3 for video?	It has no Ethernet, its memory cannot hold one 2560×1440 frame, and it cannot decode H.264. So the recorder runs on the ROCK 4C+.
Does it work with a real gecko?	Not tested yet. Every test used a toy lizard; the real enclosure is our first next step.
Are the images on slides 08, 10, 12 and 14 live?	No. They are replays: recorded clips fed back through the same detector and tracker, with boxes drawn on each frame.
What caused the 31.4 % change?	It is not visible in the recording; the rule treated it as a scene change and ignored it.

Facts from project-camera/experiments (30 Oct and 5 Nov 2026). Numbers match the Figma “Final PPT” slides 05–15.